Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 70%. We keep your highest score.

Next item →

1. What is the coefficient of x^3y^4 in the expansion of $(3x-2y)^7$? 1 / 1 point

- $\binom{7}{4} \times 6^3$
- $\binom{7}{3}$
- $\binom{7}{4} \times (-6)^3$
- $\binom{7}{3} \times 3^3 \times (-2)^4$

Nice work

2. Which one is equal to $\sum_{r=0}^{n} 2^r\ C(n,r)$? 1 / 1 point

- 9^n
- 3^n
- 4^n
- 2^n

Nice work

Note that $(x+y)^n = \sum_{r=0}^{n} C(n,r) \cdot x^r \cdot y^{n-r}$. If we substitute $x=1, y=2$ we will have RHS as $\sum_{r=0}^{n} 2^r\ C(n,3)$ which will be equal to 3^n.

3. What is the coefficient of x^2y^4 in the expansion of $(x+y)^6$? 1 / 1 point

- 16
- 20
- 15
- 12

Nice work

The answer is $\binom{6}{2} = 15$.

4. Consider the expansion of $(\sqrt[4]{a^3} + \frac{1}{\sqrt[3]{a^2}})^{17}$. What is the independent term from a? (In other words what is the constant value in the expansion) 1 / 1 point

- $\binom{17}{12}$
- $\binom{17}{7}$
- $\binom{17}{9}$
- $\binom{17}{11}$

Nice work

Notice that $(\sqrt[4]{a^3} + \frac{1}{\sqrt[3]{a^2}})^{17} = \sum_{r=0}^{17} \binom{17}{3} (a^{\frac{3}{4}})^{17-r} (a^{\frac{-2}{3}})^r$ which is equal to Extra open brace or missing close brace. In order to have the independent term we must have $\frac{51-3r}{4} - \frac{2r}{3} = 0 \implies r = 9$. So the independent term is equal to $\binom{17}{9}$.

5. Which one is equal to $C(n,1) + C(n,2) + \cdots + C(n,n-1)$? 1 / 1 point

- $1+2+4+\cdots+2^{n-1}$
- 2^n
- $2(2^{n-1}-1)$
- $1+2+4+\cdots+2^n$

Nice work

Note that $C(n,0) + C(n,1) + \cdots + C(n,n) = 2^n$. The statement in the question is missing two terms that are $C(n,0) = 1$ and $C(n,n) = 1$. So $C(n,1) + C(n,2) + \cdots + C(n,n-1) = 2^n - 2 = 2(2^{n-1}-1)$.

6. How many different term exists in the expansion of $(a+b+c+d)^{10}$? 1 / 1 point

- $\binom{14}{4}$
- $\binom{13}{4}$
- $\binom{13}{3}$
- $\binom{14}{3}$

Nice work

Note that if we expand this relation we will have terms of form $a^x b^y c^t d^r$ such that $x+y+t+r = 10$. We only need to find the number of non-negative solutions to this equation.

7. What is the number of rational terms in the expansion of $(\sqrt{2} + 3^{\frac{1}{4}})^{12}$ 1 / 1 point

- 3
- 2
- 4
- 5

Nice work

Note that to have a rational term, $n-k$ must be even in $(\sqrt{2})^{12-k}$ and k must be a multiple of 4 in $(3^{\frac{1}{4}})^k$. Note that the only possible values for k is equal to $\{0,4,8,12\}. And for any valid$k,12-k$$ will be automatically an even number so there will not be any inconsistencies.

8. $\sum_{k=0} 6\binom{52-k}{5}$ is equal to: 1 / 1 point

- $C(53,6) - C(46,6)$
- $C(53,5) - C(46,5)$
- $C(54,5) - C(46,5)$
- $C(54,6) - C(46,6)$

Nice work

We have
$C(46,5) + C(47,5) + \cdots + C(52,5) = C(46,6) + C(46,5) + C(47,5) + \cdots + C(52,5) – C(46,6)$

We also know that $C(n,k) = C(n-1,k) + C(n-1,k-1)$. So we have
$C(46,6) + C(46,5) + C(47,5) + \cdots + C(52,5) – C(46,6) = C(53,6) - C(46,6)$.